



MediCrypt – Secure Medical Consultation & Health Record Management System

1. Problem Statement

Nowadays, hospitals store patient consultation details, prescriptions, and medical records digitally. If this information is not properly secured, unauthorized users or hackers may access sensitive patient data. This can lead to privacy issues and data breaches.

The MediCrypt system aims to solve this problem by providing secure doctor-patient communication, encrypting consultation data, and protecting medical records so that only authorized users can access them.

2. Objectives

The main objectives of this project are:

- To provide secure communication between doctors and patients.
 - To encrypt consultation details before storing them.
 - To protect medical records from unauthorized access.
 - To implement secure login and role-based access.
 - To maintain patient privacy and data confidentiality.
 - To keep track of important system activities using audit logs.
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3. Scope

The MediCrypt system focuses on securing medical consultation and patient records.

The system includes:

- Patient Registration and Login
- Doctor Login
- Appointment Booking
- Secure Consultation
- Medical Record Upload
- Prescription Management
- Encrypted Data Storage
- Role-Based Access Control
- Audit Logs

The system does not include:

- Hospital Billing

- Pharmacy Management
 - Laboratory Management
 - Ambulance Services
 - Inventory Management
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4. Proposed Solution

MediCrypt is a secure web application that allows patients to consult doctors safely. Patients can book appointments, upload medical reports, and view prescriptions. Doctors can access patient records, provide consultation notes, and upload prescriptions.

All sensitive information is encrypted before it is stored in the database. This ensures that patient data remains safe and can only be accessed by authorized users.

5. Expected Outcome

After implementing the system:

- Patient information will be more secure.
- Doctor-patient communication will be protected.
- Medical records will remain confidential.
- Unauthorized users cannot access sensitive data.
- Hospitals can manage consultations more securely and efficiently.